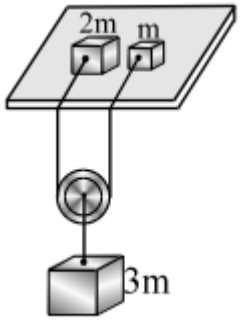
Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Two particles of masses m and $2\mathrm{m}$ are placed on a smooth horizontal table. A string, which joins them hangs over the edge supporting a light pulley, which carries a mass $3\mathrm{m}$. The two parts of the string on the table are parallel and perpendicular to the edge of the table. The parts of the string outside the table are vertical. Find the acceleration of the particle of mass $3\mathrm{m}$.



(A) $9\ \mathrm{g}/17$

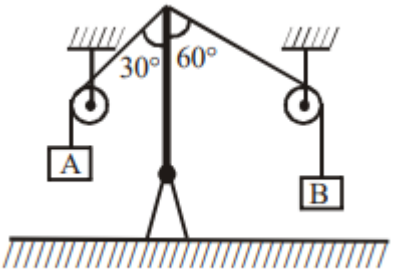
(B) $8\ \mathrm{g}/17$

(C) $7\ \mathrm{g}/17$

(D) None of these

Q.2

A rod can freely rotate in vertical plane about the hinge at its bottom. Two strings tie the top of the rod with blocks A and B as shown in the figure. At the instant shown, the rod is vertical and the speed of block A is $\mathrm{u} = 2\sqrt{3}\ \mathrm{m/s}$ downwards. Find the speed of block B at shown instant in $\mathrm{m/s}$.



(A) $6\ \mathrm{m/s}$

(B) $3\ \mathrm{m/s}$

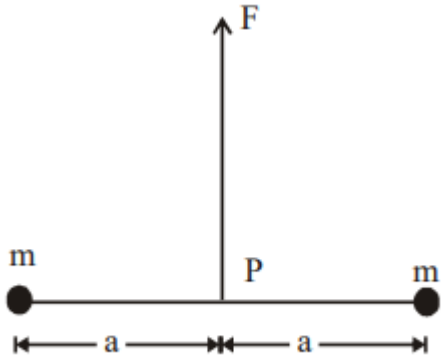
(C) $2\ \mathrm{m/s}$

(D) $1\ \mathrm{m/s}$

Q.3

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Two particles of mass $m = 1\ \text{kg}$. each are tied at the ends of a light string of length $10\ \text{cm}$. The whole system is kept on a frictionless horizontal surface with the string held tight so that each mass is at a distance $5\ \text{cm}$. from the centre P (as shown in the figure). Now, the mid-point of the string is pulled vertically upwards with a small but constant force $\text{F} = 8\ \text{N}$. As a result, the particles move towards each other on the surface. Find the magnitude of acceleration $\left(\text{ in } \text{m/s}^2\right)$, when the separation between them becomes $6\ \text{cm}$.



(A) $2\ \text{m/s}^2$

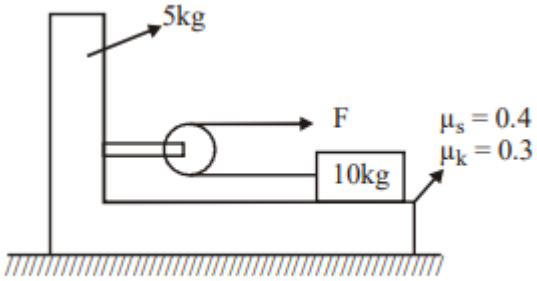
(B) $3\ \text{m/s}^2$

(C) $1\ \text{m/s}^2$

(D) $4\ \text{m/s}^2$

Q.4

A $10\ \text{kg}$ block rests on a $5\ \text{kg}$ bracket as shown in figure. The $5\ \text{kg}$ bracket rests on a horizontal frictionless surface. The coefficients of friction between the $10\ \text{kg}$ block and the bracket on which it rests are $\mu_\text{s} = 0.40$ and $\mu_\text{k} = 0.30$.



The maximum force F that can be applied if the $10\ \text{kg}$ block is not to slide on the bracket is -

(A) $32\ \text{N}$

(B) $24\ \text{N}$

(C) $18\ \text{N}$

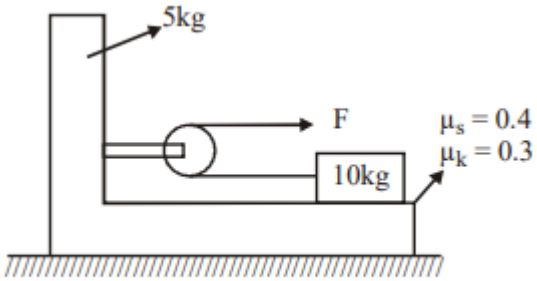
(D) $48\ \text{N}$

Q.5

A $10\ \text{kg}$ block rests on a $5\ \text{kg}$ bracket as shown in figure. The $5\ \text{kg}$ bracket rests on a horizontal frictionless surface. The coefficients of friction between the $10\ \text{kg}$ block and the bracket on which it rests are $\mu_\text{s} = 0.40$ and

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$\mu_k = 0.30$.



If $10\ \text{kg}$ block does not slide on the bracket, the corresponding acceleration of the $5\ \text{kg}$ bracket is -

(A) $1.6\ \text{m/s}^2$

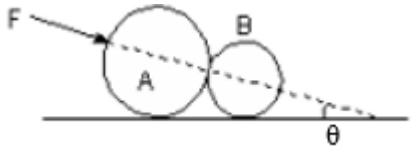
(B) $0.8\ \text{m/s}^2$

(C) $1.2\ \text{m/s}^2$

(D) $2.4\ \text{m/s}^2$

Q.6

Two spheres A and B are placed touching each other on a rough horizontal surface such that line joining their Centers make an angle θ with horizontal and along this line, a force F is applied on A. The friction is absent between spheres and coefficient of friction is μ between spheres and horizontal surface. Find the appropriate condition for which spheres are impending to slide:



(A) $\mu = \tan\theta$

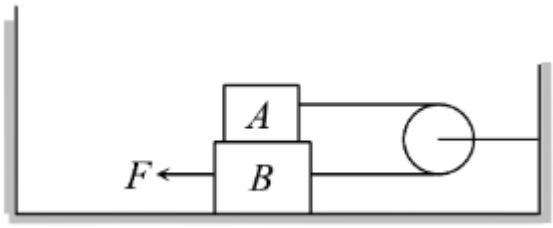
(B) $\mu \geq \tan\theta$

(C) $\mu < \cot\theta$

(D) $\mu \geq \cot\theta$

Q.7

The masses of the blocks A and B are $0.5\ \text{kg}$ and $1\ \text{kg}$ respectively. These are arranged as shown in the figure and are connected by a massless string. The coefficient of friction between all contact surfaces is 0.4 . The force needed to move the block B with constant velocity will be $\left(g = 10\ \text{m/s}^2\right)$



(A) $5\ \text{N}$

(B) $10\ \text{N}$

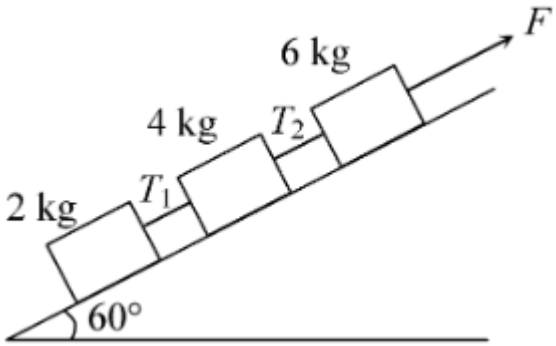
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(C) $15\ \text{N}$

(D) $20\ \text{N}$

Q.8

Three blocks of masses $2\ \text{kg}, 4\ \text{kg}$ and $6\ \text{kg}$ are connected by string and resting on a frictionless incline of 60° as shown. A force of $120\ \text{N}$ is applied upward along the incline to the $6\ \text{kg}$ block. If the strings are ideal, the ratio T_1/T_2 will be $\left(g = 10\ \text{ms}^{-2}\right)$



(A) $1:1$

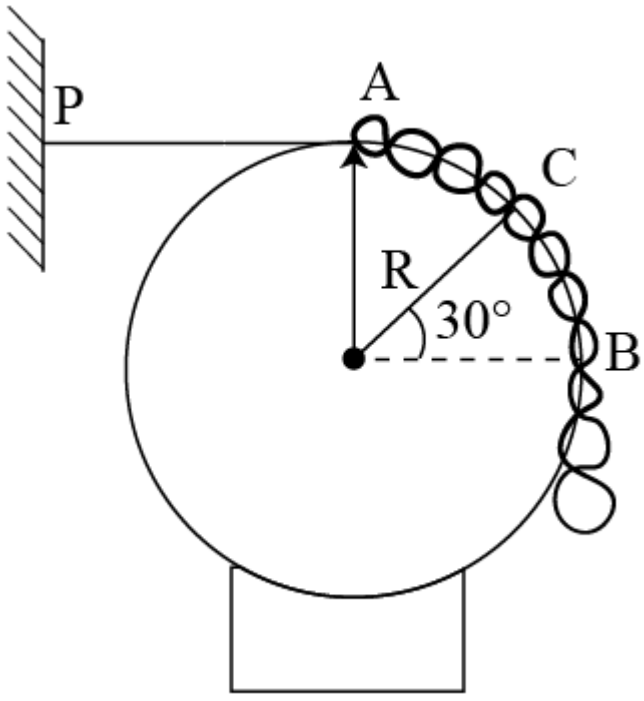
(B) $1:2$

(C) $1:3$

(D) $1:4$

Q.9

A uniform chain of length is $L = (3/2)\text{m}$ and mass M is placed on a smooth spherical surface of radius $R = \frac{2}{\pi}$ \mathrm{m}$withoneend$A\$ held by thread PA. then just after PA is burnt, the chain starts with acceleration 'a' then:



(A) Tension at point $C = \frac{M}{L}\left(L - \frac{\pi R}{2}\right) g$

(B) Tension at point A is zero

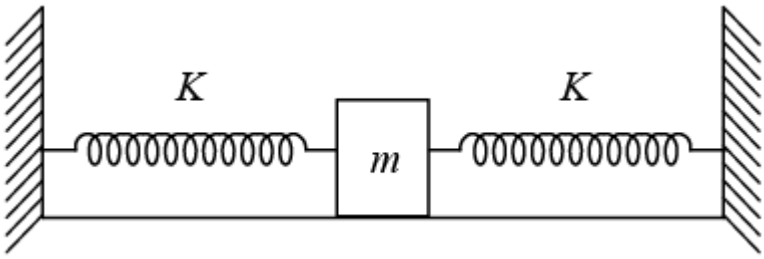
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(C) Tension at point $B = \frac{MR}{L}\left(\frac{\pi a}{2} - g\right)$

(D) Tension will increase from A to B if a $> g$

Q.10

The springs are identical and initially released. The block is pushed to left by distance x and released.



(A) Net force on block is $2kx$ just after releases.

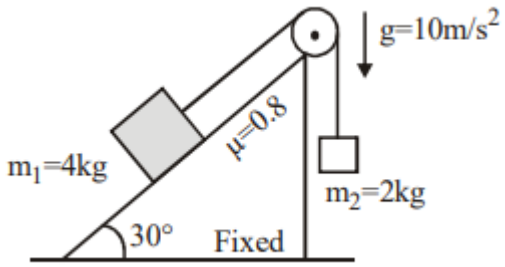
(B) Velocity of block when it reach equilibrium position is $x\sqrt{\frac{2k}{m}}$

(C) When block is at $\frac{x}{2}$ distance from equilibrium its velocity is $x\sqrt{\frac{k}{2m}}$

(D) If one spring is removed, block is pushed to left by distance x and released then velocity of block is $x\sqrt{\frac{k}{m}}$

Q.11

Two blocks of masses m_1 and m_2 are connected through a massless inextensible string. Block of mass m_1 is placed at the fixed rigid inclined surface while the block of mass m_2 hanging at the other end of the string, which is passing through a fixed massless frictionless pulley shown in figure. The coefficient of static friction between the block and the inclined plane is 0.8. The system of masses m_1 and m_2 is released from rest.



(A) The tension in the string is 20 N after releasing the system.

(B) The contact force by the inclined surface on the block is along normal to the inclined surface.

(C) The magnitude of normal force by the inclined surface on the block m_1 is $20\sqrt{3}$ N.

(D) None of these

Q.12

Passage:

Pseudo Force is a imaginary force which is recognised only by a non-Inertial observer to explain the physical situation according to Newton's Laws. Magnitude of pseudo force F_p is equal to the product of the mass m of the object and the acceleration a of the frame of reference. The direction of the force is opposite to the direction of acceleration. $F_p = -ma$

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Question: A spring weighing machine inside a stationary lifts reads $50\ \mathrm{kg}$ when a man stand on it. What would happen to the scale reading if the lift is moving upward with (i) constant velocity (ii) constant acceleration -

(A) $50\ \mathrm{kgwt}, \left(50+\frac{50\mathrm{a}}{\mathrm{g}}\right)\ \mathrm{kgwt}$

(B) $50\ \mathrm{kg}wt, \left(50+\frac{50\ \mathrm{g}}{\mathrm{a}}\right)\ \mathrm{kgwt}$

(C) $50\ \mathrm{kgwt}, \left(\frac{50\mathrm{a}}{\mathrm{g}}\right)\ \mathrm{kgwt}$

(D) $50\ \mathrm{kgwt}, \left(\frac{50\ \mathrm{g}}{\mathrm{a}}\right)\ \mathrm{kgwt}$

Q.13

Passage:
Pseudo Force is a imaginary force which is recognised only by a non-Inertial observer to explain the physical situation according to Newton's Laws. Magnitude of pseudo force $\mathrm{F_p}$ is equal to the product of the mass m of the object and the acceleration a of the frame of reference. The direction of the force is opposite to the direction of acceleration. $\mathrm{F_p} = -\mathrm{ma}$
Question: A $25\ \mathrm{kg}$ lift is supported by a cable. The acceleration of the lift when the tension in the cable is 175 N, will be -

(A) $-2.8\ \mathrm{m/s^2}$

(B) $16.8\ \mathrm{m/s^2}$

(C) $-9.8\ \mathrm{m/s^2}$

(D) $14\ \mathrm{m/s^2}$

Q.14

Passage:
Pseudo Force is a imaginary force which is recognised only by a non-Inertial observer to explain the physical situation according to Newton's Laws. Magnitude of pseudo force $\mathrm{F_p}$ is equal to the product of the mass m of the object and the acceleration a of the frame of reference. The direction of the force is opposite to the direction of acceleration. $\mathrm{F_p} = -\mathrm{ma}$
Question: A body is suspended by a string from the ceiling of an elevator. It is observed that the tension in the string is doubled when the elevator is accelerated. The acceleration will be -

(A) $4.9\ \mathrm{m/s^2}$

(B) $9.8\ \mathrm{m/s^2}$

(C) $19.6\ \mathrm{m/s^2}$

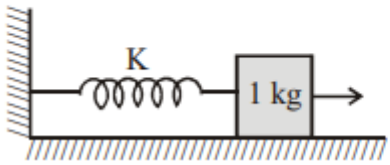
(D) $2.45\ \mathrm{m/s^2}$

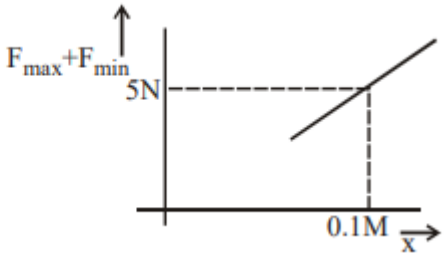
Q.15

Passage:
A block of mass $1\ \mathrm{kg}$ is placed on a rough horizontal surface. A spring is attached to the block whose other end is joined to a rigid wall, as shown in the figure. A horizontal force is applied on the block so that it remains at rest while the spring is elongated by x. $x \geq \frac{\mu \mathrm{mg}}{\mathrm{k}}$. Let $F_{\max}$ and $F_{\min}$ be the maximum and minimum values of force F

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for which the block remains a equilibrium. For a particular $\mathrm{x}, \mathrm{F}_{\max }-\mathrm{F}_{\min }=2 \mathrm{~N}$.
Also shown is the variation of $F_{\max }+F_{\min }$ versus x, the elongation of the spring.





Question: The coefficient of friction between the block and the horizontal surface is -

(A) 0.1

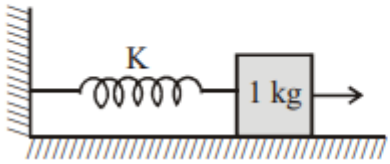
(B) 0.2

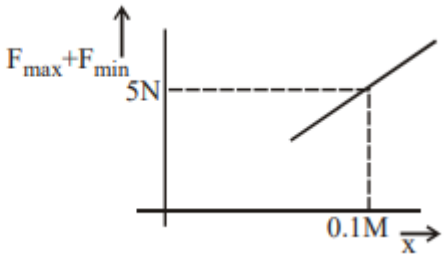
(C) 0.3

(D) 0.4

Q.16

Passage:
A block of mass $1 \mathrm{~kg}$ is placed on a rough horizontal surface. A spring is attached to the block whose other end is joined to a rigid wall, as shown in the figure. A horizontal force is applied on the block so that it remains at rest while the spring is elongated by x. $\quad x \geq \frac{\mu \mathrm{mg}}{\mathrm{k}}$. Let $F_{\max }$ and $F_{\min }$ be the maximum and minimum values of force F for which the block remains a equilibrium. For a particular $\mathrm{x}, \mathrm{F}_{\max }-\mathrm{F}_{\min }=2 \mathrm{~N}$.
Also shown is the variation of $F_{\max }+F_{\min }$ versus x, the elongation of the spring.





Question: The spring constant of the spring is -

(A) $25 \mathrm{~N} / \mathrm{m}$

(B) $20 \mathrm{~N} / \mathrm{m}$

(C) $2.5 \mathrm{~N} / \mathrm{m}$

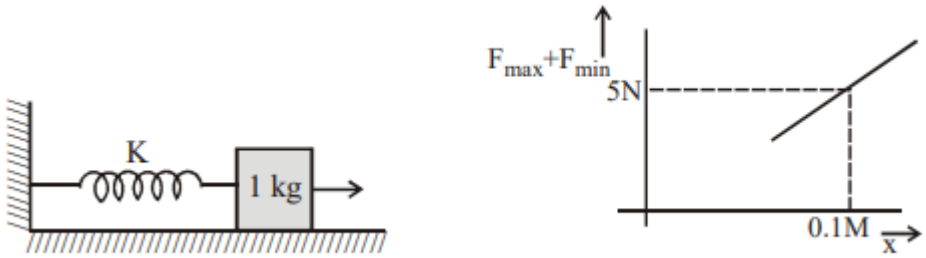
(D) $50 \mathrm{~N} / \mathrm{m}$

Q.17

Passage:
A block of mass $1 \mathrm{~kg}$ is placed on a rough horizontal surface. A spring is attached to the block whose other end is joined to a rigid wall, as shown in the figure. A horizontal force is applied on the block so that it remains at rest while the spring is elongated by x. $\quad x \geq \frac{\mu \mathrm{mg}}{\mathrm{k}}$. Let $F_{\max }$ and $F_{\min }$ be the maximum and minimum values of force F for which the block remains a equilibrium. For a particular $\mathrm{x}, \mathrm{F}_{\max }-\mathrm{F}_{\min }=2 \mathrm{~N}$.

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Also shown is the variation of $F_{max} + F_{min}$ versus x, the elongation of the spring.



Question: The value of F_{min}, if $x = 3$ cm. is $-$

(A) 0

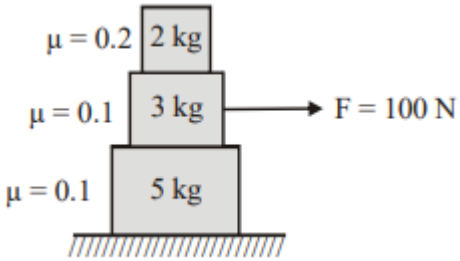
(B) 0.2 N

(C) 5 N

(D) 1 N

Q.18

For the situation shown in figure, in column I, the statements regarding friction forces are mentioned, while in column II some information related to friction forces are given. Match the entries of column I with the entries of column II.



Column I	Column II
(A) Total friction force on 3 kg block is	(p) towards right
(B) Total friction force on 5 kg block is	(q) towards left
(C) Friction force on 2 kg block due to 3 kg block is	(r) zero
(D) Friction force on 3 kg block due to 5 kg block is	(s) non-zero

(A) A-(qs) B-(r) C-(ps) D-(qs)

(B) A-(qr) B-(qs) C-(rs) D-(pq)

(C) A-(pr) B-(qs) C-(ps) D-(pq)

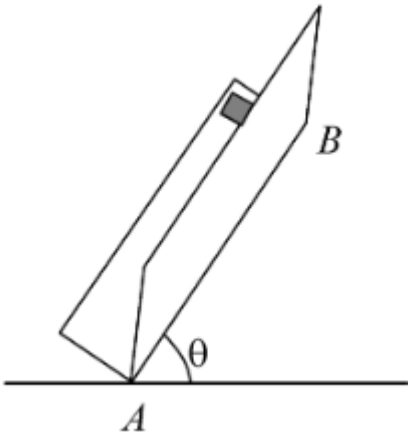
(D) A-(pq) B-(ps) C-(rs) D-(qs)

Q.19

A block of mass m is sliding on an inclined right angle through as shown in figure. If μ is the coefficient of kinetic

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friction, find the acceleration of the block is $g[\sin\theta - k\mu\cos\theta]$. Find the value of k.



Q.20

A $12.0 - \text{kg}$ box rests on the level bed of a truck. The coefficients of friction between the box and bed are $\mu_s = 0.19$ and $\mu_k = 0.15$. The truck stops at a stop sign and then starts to move with an acceleration of $2.20\ \text{m/s}^2$. If the box is $1.80\ \text{m}$ from the rear of the truck when the truck starts, how much time elapses before the box falls off the truck?

Answers & Solutions

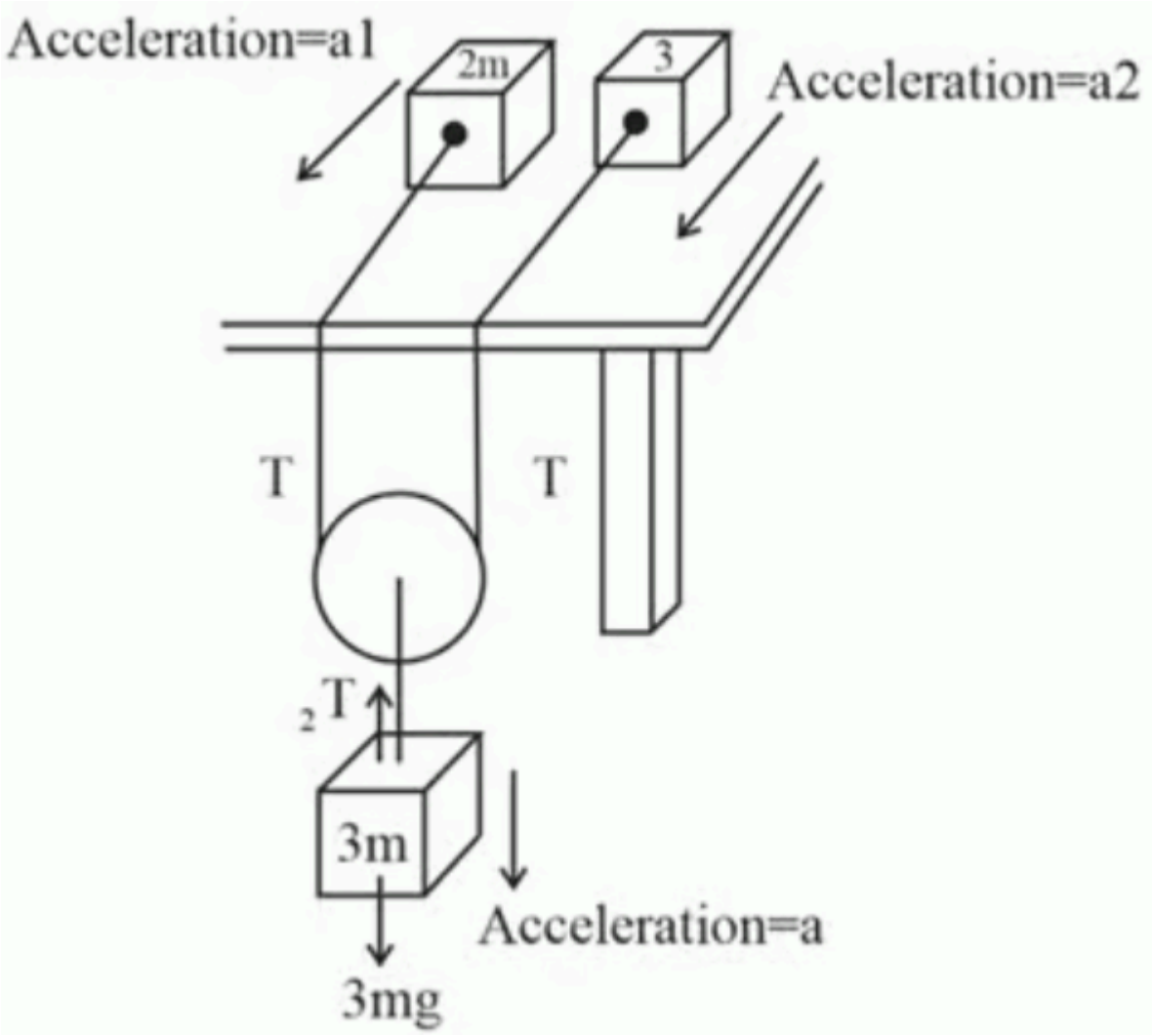
Q.1 Answer:

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Solution:

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Drawing the free body diagram of the given situation having tension T in the strings



Using constraint relation, we do find the relation between the acceleration of all masses.

Let a is the acceleration of mass $3\ \mathrm{m}$

a_1 is the acceleration of $2m$

a_2 is the acceleration of m

From the free body equation for the bodies,

$T = 2ma_1 \quad \ldots 1$

$T = ma_2 \quad \ldots 2$

Q.2 Answer:

$6\ \mathrm{m/s}$

Solution:

Let the magnitude of velocity of top of the rod be v_R towards left and the speed of block B be v.

$\ell_1 + \ell_2 =$ **constant**

$\therefore \quad \frac{\mathrm{d}\ell_1}{\mathrm{dt}} + \frac{\mathrm{d}\ell_2}{\mathrm{dt}} = 0$

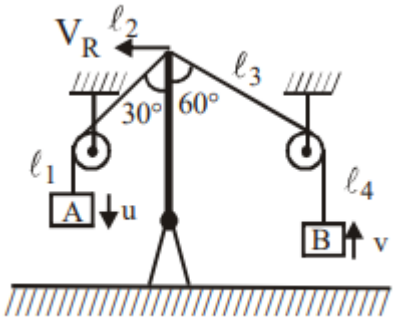
$\Rightarrow \mathrm{u} - \mathrm{v_R} \cos 60^\circ = 0 \ldots (i)$

$\ell_3 + \ell_4 =$ **constant**

$\therefore \quad \frac{\mathrm{d}\ell_3}{\mathrm{dt}} + \frac{\mathrm{d}\ell_4}{\mathrm{dt}} = 0 \Rightarrow \mathrm{v_R} \cos 30^\circ - \mathrm{v} = 0 \ldots (ii)$

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Solving eq. (1) and (2), $v = \sqrt{3}u = 6\ m/s$



Q.3 Answer:

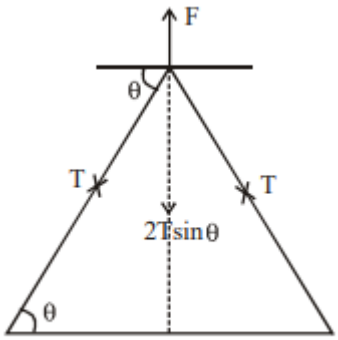
$3\ m/s^2$

Solution:

$F = 2\ T \sin\theta$

$a = \frac{T\cos\theta}{m}$

$a = \frac{F\cos\theta}{2\ m\sin\theta} = \frac{8}{2\times1}\frac{3}{\sqrt{5^2-3^2}} = 3\ m/s^2$



Q.4 Answer:

$24\ N$

Solution:

Assuming that the system moves together and there is no sliding, therefore, acceleration of the system

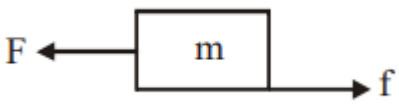
$a = \frac{F}{(5+10)}; a = \frac{F}{15} \ldots(1)$

$f - F = ma = 10\left(\frac{F}{15}\right); \quad f = F\left(1 + \frac{10}{15}\right) \ldots$**(2)**

If there is no sliding $F \leq \mu_s N$

$F\left[\frac{5}{3}\right] \leq 0.4 \times 10 \times 10 \Rightarrow F = 24\ N$

From eq. (1), $a = \frac{F}{15} = \frac{24}{15} = 1.6\ m/s^2$



Q.5 Answer:

$1.6\ m/s^2$

Solution:

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Assuming that the system moves together and there is no sliding, therefore, acceleration of the system

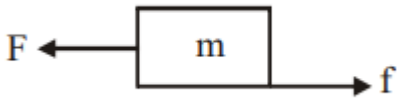
$a = \frac{F}{(5+10)}; a = \frac{F}{15} \ldots (1)$

$\mathrm{f} - \mathrm{F} = \mathrm{ma} = 10\left(\frac{\mathrm{F}}{15}\right); \quad \mathrm{f} = \mathrm{F}\left(1 + \frac{10}{15}\right) \ldots$**(2)**

If there is no sliding $\mathrm{F} \leq \mu_s \mathrm{N}$

$\mathrm{F}\left[\frac{5}{3}\right] \leq 0.4 \times 10 \times 10 \Rightarrow \mathrm{F} = 24\ \mathrm{N}$

From eq. (1), $\mathrm{a} = \frac{\mathrm{F}}{15} = \frac{24}{15} = 1.6\ \mathrm{m/s^2}$



Q.6 Answer:

$\mu < \cot\theta$

Solution:

Along horizontal,

$\mathrm{F}\cos\theta = \mathrm{R1}\cos\theta + \mu N_1$

Along vertical

$\mathrm{F}\sin\theta_{+}\mathrm{mg} = \mathrm{N1} + \mathrm{R1}\sin\theta$

$\text{or } R_1 = \dfrac{F\sin\theta + mg - N_1}{\sin\theta}$

From eqn. (1)

$\mathrm{F}\cos\theta = \left(\dfrac{F\sin\theta + mg - N_1}{\sin\theta}\right)\cos\theta + \mu N_1$

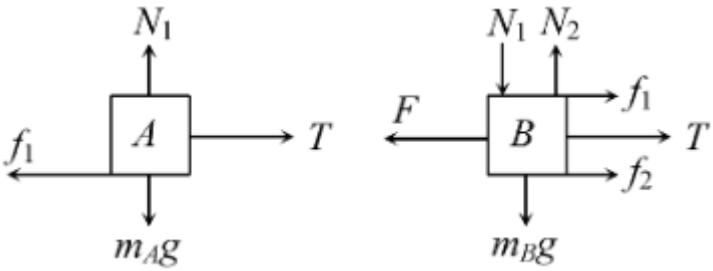
$= F\cos\theta + mg\cot\theta - N_1\cot\theta + \mu N_1$

$N_1 = \dfrac{mg\cot\theta}{(\cot\theta - \mu)} \quad N_1 > 0 \Rightarrow \mu < \cot\theta$

Q.7 Answer:

10 N

Solution:



$m_A = 0.5\ \mathrm{kg}; \quad m_B = 1\ \mathrm{kg}$

From F.B.D. of block A**,**

$T = f_1 = \mu m_A g = 2N$

From F.B.D. of block B**,**

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$f_2 = \mu N_2 = 6N$

$F = T + f_1 + f_2 = 2 + 2 + 6 = 10N$

$\therefore$ **(b)**

Q.8 Answer:

$1 : 3$

Solution:

$m_1 = 2\ \text{kg}, m_2 = 4\ \text{kg}, m_3 = 6\ \text{kg}, \quad a = \dfrac{F - (m_1 + m_2 + m_3)\, g \sin 53^\circ}{m_1 + m_2 + m_3}$

$a = \dfrac{120 - 12 \times 10 \times 4/5}{12}, \quad = \dfrac{24}{12} = 2\ \text{ms}^{-2}$

$T_1 - m_1 g \sin 53^\circ = m_1 a, \quad T_1 = 4 + 20 \times \dfrac{4}{5} = 20N$

$T_2 - (m_1 + m_2)\, g \sin 53^\circ = (m_1 + m_2)\, a, \quad T_2 = 12 + 60 \times \dfrac{4}{5} = 60\ \text{N}, \quad \dfrac{T_1}{T_2} = \dfrac{1}{3}$

$\therefore \quad (\text{c})$

Q.9 Answer:

Tension at point A is zero

Solution:

$T + dT + dmg \sin\theta - T = dna$

$dT + dug \sin\theta = dua$

$dw = \dfrac{M}{L} dx = \dfrac{M}{L} R d\theta$

$\therefore \int_0^J dT + \dfrac{M}{L} gR \int \sin\theta d\theta = \dfrac{M}{L} Ra \int_0^\pi d\theta$

$(\text{at end A, T} = 0\ )$

$\Rightarrow \quad T = \dfrac{MR}{L}[a\theta - g](1 - \cos\theta)$

For T_b put $\theta = \pi/2$

$\therefore \text{T}_\text{b} = \frac{MR}{L}\left(\frac{\pi a}{2} - g\right)$

Q.10 Answer:

Net force on block is 2kx just after releases.

Solution:

$W_s = \Delta k$

$2\left(\frac{1}{2}kx^2\right) = \frac{1}{2}mv^2 \quad \longrightarrow 2kx$

$v = x\sqrt{\frac{2k}{m}}$

$W_s = \Delta k$

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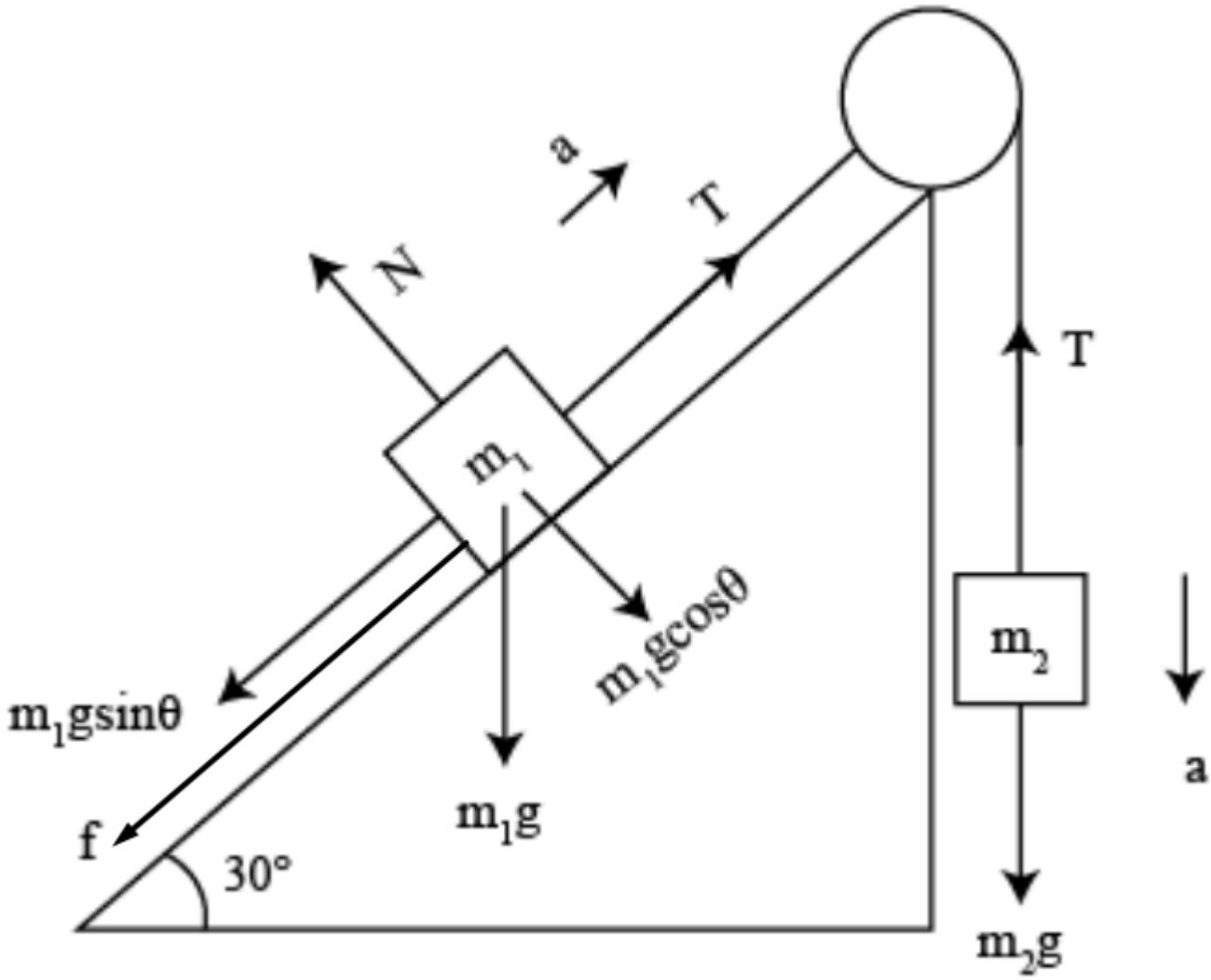
$\frac{1}{2}kx^2 = \frac{1}{2}mv^2$

$v = x\sqrt{\frac{k}{m}}$

Q.11 Answer:

The tension in the string is 20 N after releasing the system.

Solution:



Given:- $\mu = 0.8$

The system has $a = 0$ **just after releasing from rest For** $m_2 \rightarrow$

$m_2g = T = m_2a$

$a = 0$

$\Rightarrow T = m_2g = 20$ N

for $m_1 \rightarrow$

$N = m_1g\cos\theta = 4 \times 10\cos 30$

$N = 20\sqrt{3}N$ **this is perpendicular to inclined surface.**

correct option $(1, 2, 3)$

Q.12 Answer:

50 kgwt, $\left(50 + \frac{50a}{g}\right)$ kgwt

Solution:

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(i) In the case of constant velocity of lift, there is no reaction,
therefore the apparent weight = actual weight. Hence the reading of machine is 50 **kgwt.**
(ii) In this case the acceleration is upward the reaction $R =$ **ma acts downward, therefore apparent weight is more than actual weight i.e.** $\mathrm{W}' = \mathrm{W} + \mathrm{R} = \mathrm{m}(\mathrm{g} + \mathrm{a})$
Hence, scale show a reading of $\mathrm{m}(\mathrm{g} + \mathrm{a})$
$= (50 + 50a/g)\mathrm{kg\ wt}$

Q.13 Answer:

$-2.8\ \mathrm{m/s^2}$

Solution:

Tension $= \mathrm{m}(\mathrm{g} + \mathrm{a})$**, when lift moving up, putting the values, we get** $175 = 25(9.8 + \mathrm{a}) \Rightarrow \mathrm{a} = -2.8\ \mathrm{m/s^2}$ **[negative sign shows that lift is moving downward]**

Q.14 Answer:

$9.8\ \mathrm{m/s^2}$

Solution:

Apparent tension, $\mathrm{T} = 2\ \mathrm{T_0}$
So, $\mathrm{T} = 2\ \mathrm{T_0} = \mathrm{T_0}\left(1 + \frac{\mathrm{a_0}}{\mathrm{g}}\right)$ **or** $2 = 1 + \frac{\mathrm{a_0}}{\mathrm{g}} \Rightarrow \mathrm{a_0} = \mathrm{g} = 9.8\ \mathrm{m/s^2}$

Q.15 Answer:

0.1

Solution:

$$F_{\max} = \mathrm{kx} + \mu\mathrm{mg}$$
$$F_{\min} = \mathrm{kx} - \mu\mathrm{mg}$$
$$\therefore \quad F_{\max} + F_{\min} = 2\mu\mathrm{mg} \text{ or } 2 = 2\mu 10$$
$$\therefore \quad \mu = 0.1$$

When $\mathrm{x} = 0.03$
$\mathrm{kx} = 25 \times 0.03 = 0.75\ \mathrm{N}$**, which is less than** $\mu\mathrm{mg} = 0.1 \times 10 = 1\ \mathrm{N}$
$\therefore$ **The block will be at rest, without applying force F.**

Q.16 Answer:

$25\ \mathrm{N/m}$

Solution:

$$F_{\max} = \mathrm{kx} + \mu\mathrm{mg}$$
$$F_{\min} = \mathrm{kx} - \mu\mathrm{mg}$$
$$\therefore \quad F_{\max} + F_{\min} = 2\mu\mathrm{mg} \quad \text{or } 2 = 2\mu 10$$
$$\therefore \quad \mu = 0.1$$
$$F_{\max} + F_{\min} = 2\mathrm{kx} \ldots (1)$$

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Putting in eq. (1)

$t = 2k(0.1); k = 25\ N/m$

When $x = 0.03$

$kx = 25 \times 0.03 = 0.75\ N$, **which is less than** $\mu mg = 0.1 \times 10 = 1\ N$

$\therefore$ **The block will be at rest, without applying force F.**

Q.17 Answer:

0

Solution:

$F_{max} = kx + \mu mg$

$F_{min} = kx - \mu mg$

$\therefore \quad F_{max} + F_{min} = 2\mu mg \quad$ or $2 = 2\mu 10$

$\therefore \quad \mu = 0.1$

$F_{max} + F_{min} = 2kx ...(1)$

Putting in eq. (1)

$t = 2k(0.1); k = 25\ N/m$

When $x = 0.03$

$kx = 25 \times 0.03 = 0.75\ N$, **which is less than** $\mu mg = 0.1 \times 10 = 1\ N$

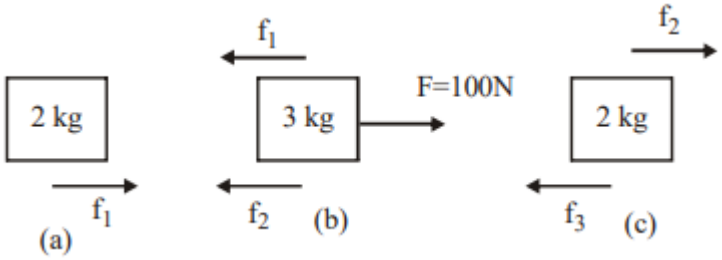
$\therefore$ **The block will be at rest, without applying force F.**

Q.18 Answer:

A-(qs) B-(r) C-(ps) D-(qs)

Solution:

(A) $-\mathbf{qs}, (\mathbf{B}) - \mathbf{r}, (\mathbf{C}) - \mathbf{ps}, (\mathbf{D}) - \mathbf{qs}$



$f_{\ell_1} = 0.2 \times 2\ g = 4\ N$

$f_{\ell_2} = 0.1 \times 5\ g = 5\ N$

$f_{\ell_3} = 0.1 \times 10\ g = 10\ N$

Friction on $3\ kg$ **block is towards left and non-zero. Hence, (A) - qs** $f_{\ell_2} < f_{\ell_3}$**, hence** $5\ kg$ **block will not move. So the net friction on** $5\ kg$ **will be zero. Hence, (B)-r**

Q.19 Answer:

—

Solution:

$R' = 2R\cos 45 = \sqrt{2}R$ **...(i)**

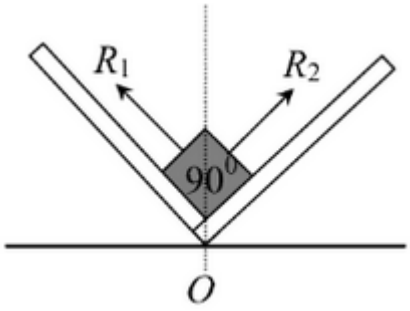
Now for equilibrium in a direction perpendicular to AB

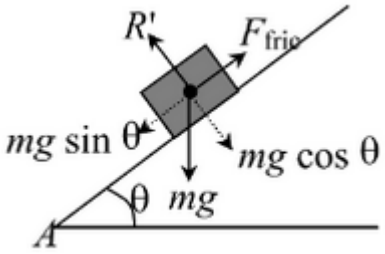
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$R' = mg\cos\theta$ **...(ii)**

which in the light of equation (i) gives

$R = (1/\sqrt{2})mg\cos\theta$ **...(iii)**





And for the motion along the trough

$mg\sin\theta - \mu R_1 - \mu R_2 = ma$

substituting R from equation (ii) in the above

$a = g\sin\theta - \frac{2\mu}{m} \times \frac{mg\cos\theta}{\sqrt{2}}$

i.e. $a = g[\sin\theta - \sqrt{2}\mu\cos\theta]$

Q.20 Answer:

—

Solution:

IDENTIFY: Apply $\Sigma\vec{F} = m\vec{a}$ to the box. Compare the acceleration of the box to the acceleration of the truck and use constant acceleration equations to describe the motion.

SET UP: Both objects have acceleration in the same direction; take this to be the $+x$-direction.

EXECUTE: If the box were to remain at rest relative to the truck, the friction force would need to cause an acceleration of $2.20\ \text{m/s}^2$; however, the maximum acceleration possible due to static friction is $(0.19)\left(9.80\ \text{m/s}^2\right) = 1.86\ \text{m/s}^2$, and so the box will move relative to the truck; the acceleration of the box would be $\mu_\text{k} g = (0.15)\left(9.80\ \text{m/s}^2\right) = 1.47\ \text{m/s}^2$. The difference between the distance the truck moves and the distance the box moves (i.e., the distance the box moves relative to the truck) will be $1.80\ \text{m}$ after a time $t = \sqrt{\frac{2\Delta x}{a_\text{truck} - a_\text{box}}} = \sqrt{\frac{2(1.80\ \text{m})}{(2.20\ \text{m/s}^2 - 1.47\ \text{m/s}^2)}} = 2.221\ \text{s}$.

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